

2024

COMPUTER SCIENCE — HONOURS — PRACTICAL

Paper : DSCC-3P

(Data Structure)

Full Marks : 25

SET - III

Marks Distribution :

Experiment : 15

Source Code : 08

Algorithm : 04

Output : 03

Viva voce : 05

Laboratory Notebook : 05

Answer *any one* question.

1. Write a program in C to perform the following operations on an array of integers :
 - (a) Take Command line inputs to initialize the array.
 - (b) Display the content of the array using a function.
 - (c) Write a function to arrange the data in ascending order using Quick Sort Technique.

2. Write a program in C to perform the following operations on an array of integers :
 - (a) Create the data file first. Initialize the array from file. Take the file name from Command line.
 - (b) Display the content of the array using a function.
 - (c) Write a function to search 2nd minimum value from the array without ordering the array.

3. Write a program in C to perform the following operations in a singly linked list with the node structure {Data, Next}.
- (a) Take Command line inputs to initialize the linked list.
 - (b) Display the content of the linked list using a function.
 - (c) Write a function to delete the maximum value node from the linked list.
 - (d) Write a function to reverse the linked list.
4. Write a menu driven program in C to perform the following operations in a Stack using linked list with the node structure {Roll, Name, Next}.
- (a) Push
 - (b) Pop
 - (c) Display
 - (d) Reverse.
5. Write a menu driven program in C to perform the following operations in a queue using the concept of circular array. Assume the contents are integer.
- (a) Insert
 - (b) Delete
 - (c) Display.
6. Write a menu driven program in C to perform the following operations on Matrix. Assume the contents are integer. Check the necessary condition for performing the addition and multiplication operations.
- (a) Initialize
 - (b) Addition
 - (c) Multiplication.
7. Write a menu driven program in C to perform the following operations in a Heap Tree(Max). Assume the contents are integer.
- (a) Insert
 - (b) Delete
 - (c) Display (as array).
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